7.5. Double Circulation

Circulation of the blood was first described by **William Harvey** (1628). There are two types of blood circulation in vertebrates, **single circulation** and **double circulation** which is shown in Figure 7.10 (a and b) and 7.11.

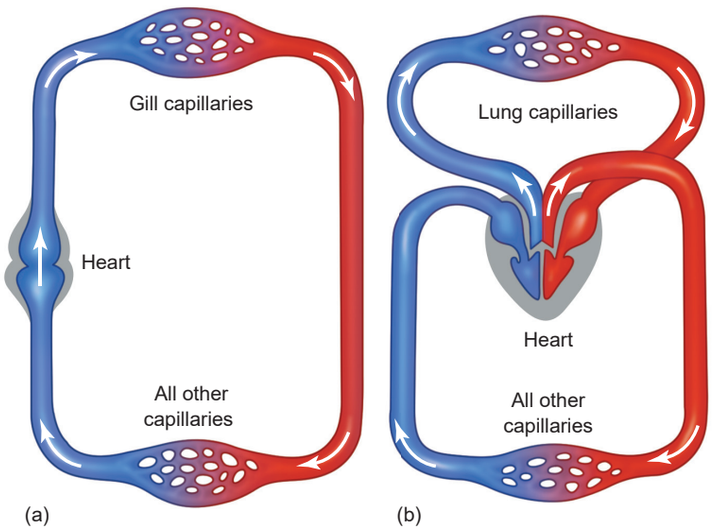


Figure 7.10 Diagrammatic representation of (a) single circulation (b) double circulation

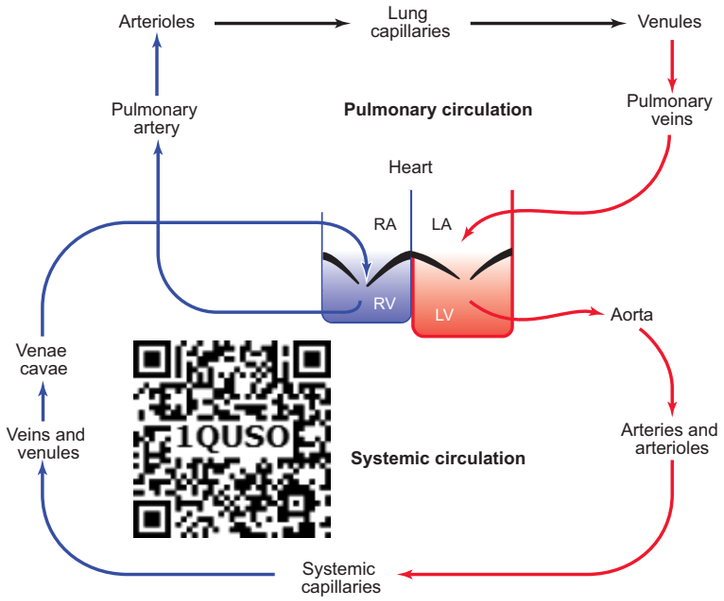


Figure 7.11 Diagrammatic representation of the Double Circulation

The blood circulates twice through the heart first on the right side then on the left side to complete one cardiac cycle. The complete double blood circulation is more prominent in mammals because of the complete partition of all the chambers (Auricles and ventricles) in the heart.

In **systemic circulation**, the oxygenated blood entering the aorta from the left ventricle is carried by a network of arteries, arterioles and capillaries to the tissues. The deoxygenated blood from the tissue is collected by venules, veins and vena cava and emptied into the right atrium. In **pulmonary circulation,** the blood from heart (right ventricle) is taken to the lungs by pulmonary artery and the oxygenated blood from the lungs is emptied into the left auricle by the pulmonary vein.

Completely separated circuits have an important advantage. Different pressures are maintained in the pulmonary and systemic circulation. Why is this advantageous? In the lungs the capillaries must be very thin to allow gas exchange, but if the blood flows through these thin capillaries under high pressure the fluid can leak through or ruptures the capillary walls and can accumulate in the tissues. This increases the diffusion distance and reduces the efficiency of the gas exchange. In contrast high pressure is required to force blood through the long systemic circuits. Hence the arteries close to the heart have increased pressure than the arteries away from the heart. Completely separated circuits (pulmonary and systemic) allow these two different demands to be met with.

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